

Accelerating the discovery of multicatalytic cooperativity

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Cooperative catalysis, in which multiple catalytic units operate synergistically, underpins a variety of synthetically and mechanistically important organic reactions^{1–4}. Despite its potential utility in new reactivity contexts, approaches to the discovery of cooperative catalysts have been limited, typically relying on serendipity or on previous knowledge of single-catalyst reactivity^{1–5}. Systematic searches for unanticipated types of catalyst cooperativity must contend with vast combinatorial complexity and are therefore not undertaken^{6–10}. Here we describe a pooling–deconvolution algorithm, inspired by group testing¹¹, which identifies cooperative catalyst behaviours with low experimental cost while accommodating potential inhibitory effects between catalyst candidates. The workflow was validated first on simulated cooperativity data and then by experimentally identifying previously documented cooperativity between organocatalysts in an enantioselective oxetane-opening reaction. The workflow was then applied in a discovery context to a Pd-catalysed decarbonylative cross-coupling reaction, enabling the identification of several ligand pairs that promote the target transformation at substantially lower catalyst loading and temperature than previously reported with single-ligand systems.

Multicatalysis, which relies on catalytic cooperativity, has proven to be powerful for promoting efficient and selective chemical transformations^{1–4}. Notably, enzymes are understood to achieve high levels of reactivity and selectivity by incorporating active-site residues that can operate cooperatively^{12–15}. Despite the enormous potential of multicatalysis, the methods by which it can be discovered remain very limited. For instance, mechanistic studies of single-catalyst reactions have led to unexpected discoveries of multicatalytic homocooperativity, in which multiple units of the same catalyst perform different but complementary roles (Fig. 1a, left). Those insights ultimately enabled the development of linked-dimer catalysts that exhibit vastly superior reactivity and selectivity relative to their monomeric counterparts^{16–21}. Multicatalytic heterocooperativity, in which different catalysts can be tailored to complementary roles, is in principle more broadly applicable but far less likely to be uncovered serendipitously, as it requires the intentional combination of multiple catalysts that have been identified to possess known complementary functions^{1,5}. Although this hypothesis-driven approach has enabled several landmark advances in synthetic catalysis^{2,3,22–24} (Fig. 1a, right), it leaves unexplored a potentially enormous reaction space of unanticipated synergistic effects resulting from previously unknown cooperativity pathways.

The primary obstacle to the empirical exploration for multicatalytic heterocooperativity is that it requires combing through a vast combinatorial catalyst space^{6–10}. For instance, a set of 50 individual candidates entails a search space of 1,225 possible pairs and 19,600 possible triples. This plethora of combinations could offer broad possibilities for discovering unexpected cooperativity if the combinatorial space can be searched thoroughly and accurately. Conventional screening

approaches, such as one-factor-at-a-time and design of experiments, have limited utility because the cooperativity landscape is likely to be highly discontinuous²⁵. Although human and machine model-based approaches are potentially more adaptable to discontinuous response surfaces, they require large sets of positive and negative training data as well as meaningful featurization and representation of cooperative effects^{26–30}, which are not generally available yet for cooperative catalytic systems.

We predicted a mechanistically agnostic yet efficient strategy for the discovery of multicatalytic heterocooperativity, in which multiple catalyst candidates are pooled and tested together. This strategy was inspired by group testing, which is used in diverse fields when limited diagnostic tests are available and when most individual tests are expected to return negative¹¹ (Fig. 1b, left). For example, group testing was applied widely during the COVID-19 pandemic³¹ to identify positive cases within a large population by testing pools of samples rather than every individual. Group testing has also been used to accelerate reaction condition optimization and discovery^{10,32–36}, but its application to the discovery of multicatalytic heterocooperativity involves a fundamental difference in that the goal is identifying positive interactions between candidates, rather than assessing individual performance. Thus, a custom deconvolution step is required to identify positive cooperative interactions from the reaction performance of pools (Fig. 1b, right), instead of simply separating individuals from a positive pool and testing them. Another key challenge is the possibility of negative cooperativity, in which antagonistic interactions between candidates mask positive cooperative signals in the same pool, affecting the sensitivity of the workflow to positive cooperativity.

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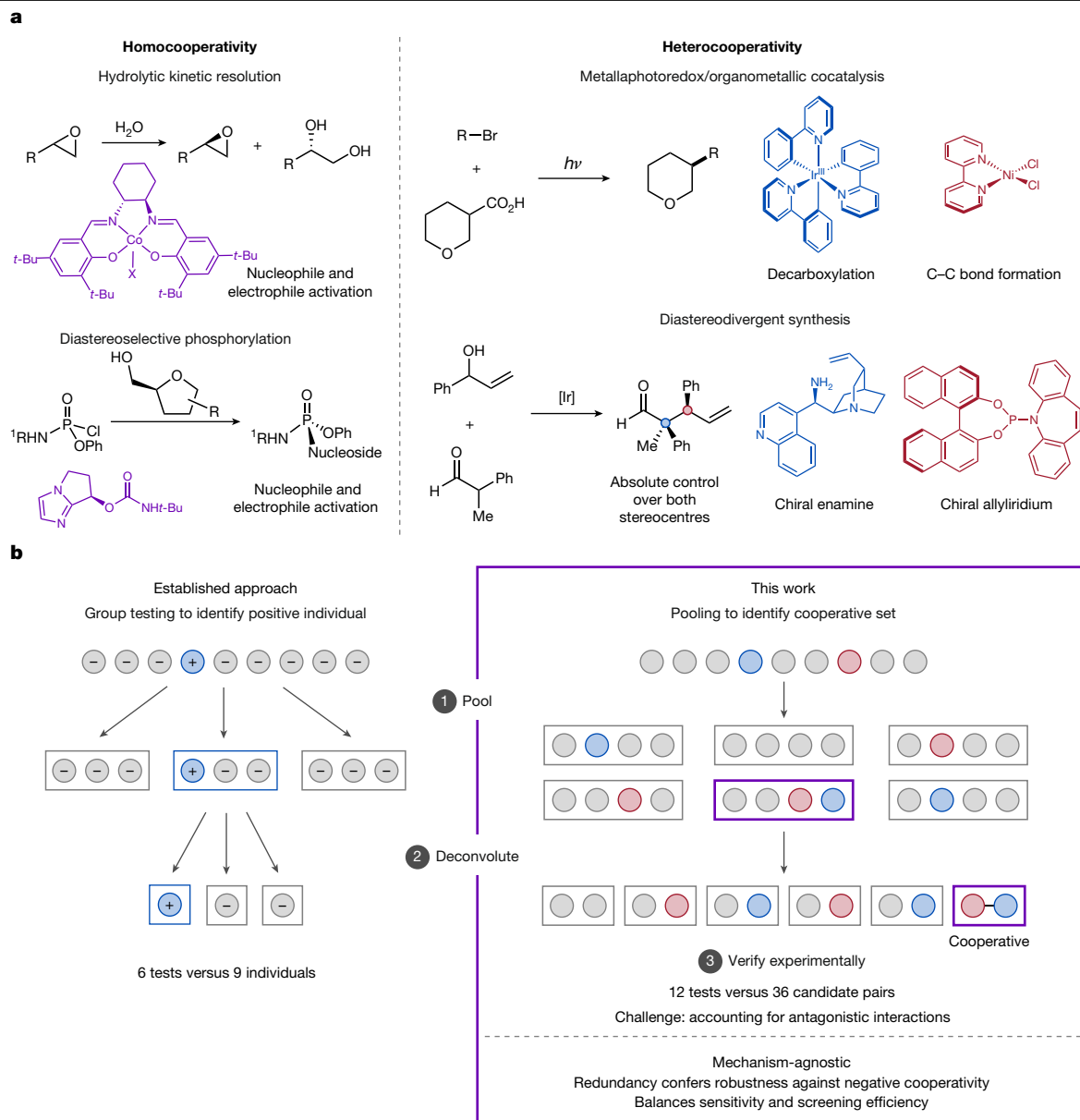


Fig. 1 | Challenges and opportunities for the empirical discovery of catalytic cooperativity. a, Seminal synthetic advances enabled by catalytic cooperativity: homocooperativity in the hydrolytic kinetic resolution, heterocooperativity in metallaphotoredox catalysis and diastereodivergent synthesis. **b**, Group testing

A simple strategy for minimizing the effects of negative cooperativity is to evaluate smaller pools, but that approach necessarily comes at experimental expense. However, by constructing specific overlapping pools designed to guarantee that catalysts meet in multiple contexts, positive cooperative interactions can be deconvoluted from the performances of pools in a more efficient manner (see below). The combination of pooling and deconvolution is crucial for approaching the optimal tradeoff between sensitivity (that is, the likelihood of finding every cooperative system within a set of catalyst candidates) and screening efficiency.

Development of a pooling–deconvolution algorithm

In our approach to identify positive cooperativity from within a set of N catalyst candidates, we set up pools with k candidates such that every subgroup of t catalysts appears in at least one pool^{37,38} (Fig. 2a, top left). The problem of assigning candidates to pools is directly related to the

as a conceptual foundation for catalyst pooling and this proposed workflow incorporating pooling and deconvolution steps to accelerate the empirical discovery of catalytic cooperativity.

mathematical concept of covering designs^{39–41}. Although covering designs generalize to cooperative sets of any size, our discussion here will focus exclusively on pairs.

An important challenge presented to a screening-for-cooperativity effort is the problem of catalyst interference, that is, the likelihood that certain candidates inhibit the performance of other pool members or pairs and thereby lead to missed hits. To accommodate that scenario, we extend covering designs to require that each candidate pair appear in multiple (specifically r , where $r \geq 1$) pools. Redundant sets of pools can be generated by randomly permuting the catalyst panel and repeating the pooling design (Fig. 2a, top middle). Thus, effective pool design must balance robustness against experimental efficiency: larger pools increase the risk of inhibitory catalysts masking cooperative pairs, whereas smaller pools, together with designed redundancy, require more experiments overall.

To quantify cooperativity between catalyst candidates within a pool, we define a cooperativity score Q derived from the performance of the

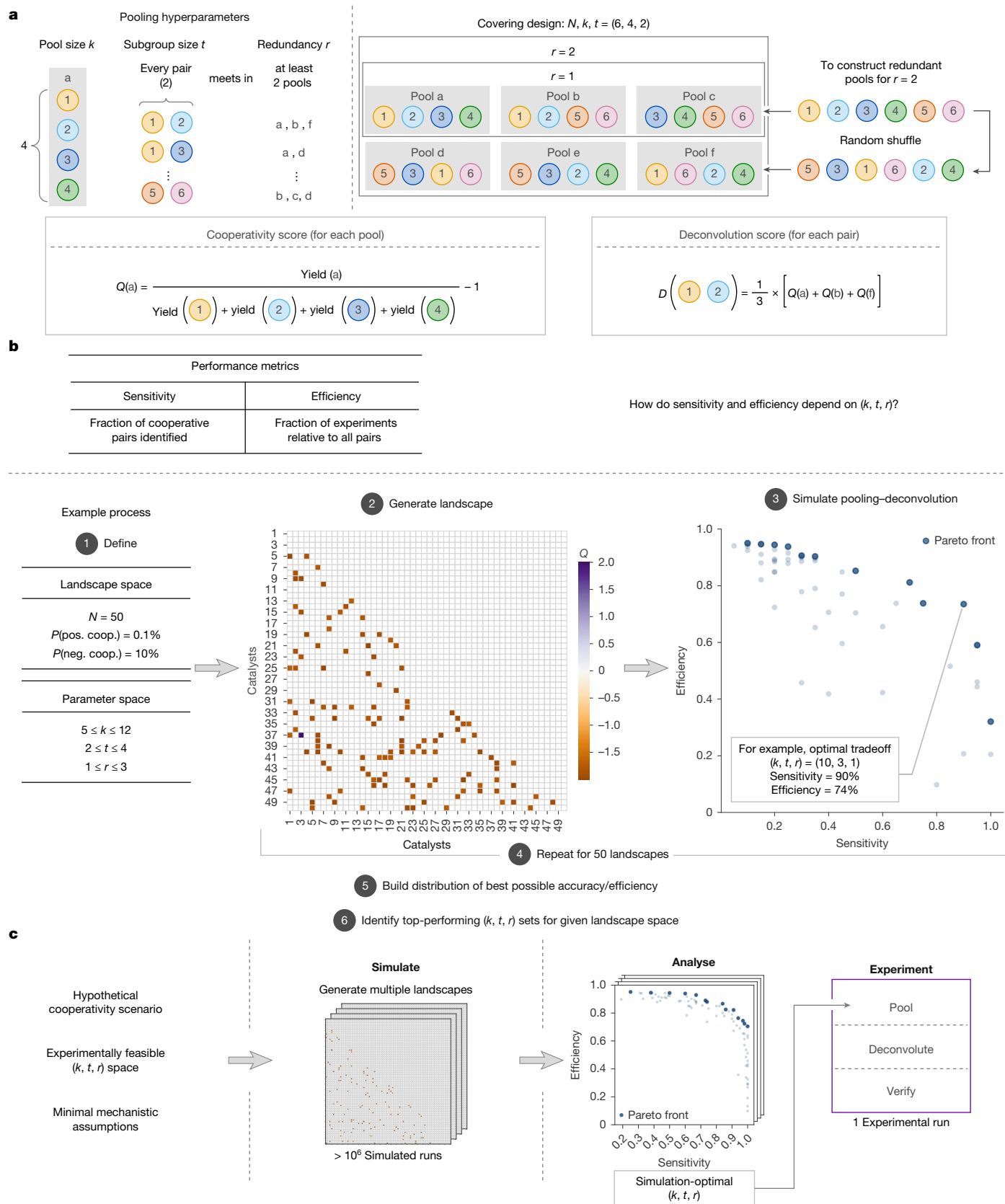


Fig. 2 | Development and simulation-based testing of pooling-deconvolution algorithm. **a**, Covering designs are the mathematical basis for the pooling step. Relevant parameters k , t and r that serve to define the optimal size and composition of the candidate pools. A cooperativity score Q can be determined for each pool, whereas a deconvolution score D can be calculated for each pair. **b**, Abstracted visualization of the computational pipeline to simulate catalytic

cooperativity and test the pooling-deconvolution workflow on various landscapes. Steps 2 and 3 are repeated over 50 randomly generated landscapes to afford statistically meaningful average values of sensitivity and efficiency. The process 1–6 can be repeated with different user settings that define the prevalence of cooperativity on hypothetical landscapes. **c**, Selection of optimal pooling parameters for experimental execution based on simulation results.

pooled catalysts relative to that of the individual catalysts. Although cooperativity can lead to enhancements in rate and/or in selectivity, we chose rate enhancement as our primary discovery objective. Pooling–deconvolution approaches could be applied in principle to the discovery of enantioselectivity enhancement by cooperative mechanisms. However, these effects could be detected only if coupled to rate enhancements as well, as highly selective but poorly active catalyst combinations may be obscured in a pool. For experimental simplicity, we select product yield at low conversion as an approximation for the initial rate. The yield of the pool (y_{pool}) is then compared with the zero-cooperativity yield expected if the candidates were not interacting ($y_{\text{composite}}$), which can be the sum, average, maximum or some other function of the yields of the individual candidates, depending on the experimental scenario. The cooperativity score Q can then be defined for a pool containing j candidates (equation (1); see Supplementary Information section 2.1 for a definition that is robust against experimental noise):

$$Q(1, 2, \dots, j) = \frac{y_{\text{pool}}(1, 2, \dots, j)}{y_{\text{composite}}(1, 2, \dots, j)} - 1 \quad (1)$$

Based on this definition, $Q > 0$ constitutes evidence for positive inter-catalyst cooperativity.

Assuming that catalyst cooperativity is relatively rare, a highly cooperative pair of candidates is expected to consistently appear in pools with high Q scores. Therefore, to estimate the performance of any pair, we can define a deconvolution score D for it by taking the average Q of all the pools that contain the pair (equation (2) and Fig. 2a, bottom). Formally, for a pair of candidates (i, j), D can be defined (equation (2)), where $P(i, j)$ is the set of pools that contains that pair. This definition of D provides an estimate of pairwise cooperativity while compensating for the possibility that negative cooperativity may affect some of the pools. The top-scoring pairs can then be verified experimentally in isolation (Fig. 1d, step 3).

$$D(i, j) = \frac{1}{|P_{ij}|} \sum_{k \in P(i, j)} Q_k \quad (2)$$

Simulations of the pooling–deconvolution workflow

We next sought to assess how different choices of parameters (k , t and r) affect the performance of the pooling–deconvolution workflow. Ostensibly, smaller pools with more redundancy lead to fewer confounding effects but a larger number of experiments. Conversely, larger pools with less redundancy could be highly efficient experimentally, but risk missing highly cooperative combinations due to poisoning effects. This tradeoff can be assessed quantitatively (Fig. 2b, top) in terms of sensitivity and efficiency. We define sensitivity (S) as the expected ($\mathbb{E}$) fraction of cooperative pairs identified relative to all the cooperative pairs in the landscape⁴²:

$$S = \mathbb{E} \left[\frac{n(\text{cooperative pairs identified})}{n(\text{cooperative pairs on landscape})} \right] \quad (3)$$

Efficiency ϵ is defined as the fraction of experiments saved compared with exhaustively testing all pairs or combinations (equation (4)), where n_{pools} is the number of experiments in the pooling stage and n_{verif} is the number of experiments in the verification phase, as both contribute non-negligible experimental cost. $\binom{N}{2} = \frac{N(N-1)}{2}$ is the number of pairs that can be formed from N candidates.

$$\epsilon = \mathbb{E} \left[1 - \frac{n_{\text{pools}} + n_{\text{valid}}}{\binom{N}{2}} \right] \quad (4)$$

To understand how sensitivity and efficiency trade off as the pooling parameters vary, we built an *in silico* pipeline to model catalytic cooperativity and to simulate pooling–deconvolution on various hypothetical cooperativity landscapes (Fig. 2b, see Supplementary Information section 3 for details). With a user-defined number of candidates N and statistical parameters for rate constants and cooperativity matrices, hypothetical cooperativity landscapes can be generated upon which pooling–deconvolution can be simulated with various combinations of (k , t , r).

We selected $N = 50$ for simulations, reasoning that pairwise testing of a library that size would be beyond reach of routine experimental execution (1,225 pairs). Simulations were conducted with 15 scenarios with varying prevalence of positive and negative cooperativity (0.1–32%) (see Supplementary Information section 3.5). In even highly challenging scenarios, the pooling–deconvolution workflow generated scores that were robust against confounding effects from negative cooperativity, resulting in the successful identification of cooperative interactions across diverse cooperativity landscapes. For instance, when positive cooperativity is rare (0.1%, 1 pair) and negative cooperativity is 100-fold more prevalent (10%, 123 pairs), the workflow performs well with $S = 0.63$ and $\epsilon = 0.70$, requiring only 365 reactions relative to 1,225 possible pairs. When positive cooperativity is more common (1%, 10 pairs) and negative cooperativity is still prevalent (10%, 123 pairs), the same level of performance is maintained with $S = 0.60$ and $\epsilon = 0.70$.

Based on these simulations, two hyperparameter sets were identified to provide the best starting points for pooling: (k , t , r) = (4, 2, 1) for rapid screening and (k , t , r) = (7, 4, 2) for more thorough pooling in challenging scenarios (see Supplementary Information section 3.5.1). Beyond these general recommendations, practitioners can run their own simulations using the pipeline described above (Fig. 2c) to identify optimal parameters (k , t , r) that balance sensitivity and efficiency for their specific number of candidates N .

Experimental validation of the pooling–deconvolution workflow

We next sought to validate the pooling–deconvolution workflow experimentally on a reaction for which cooperativity had previously been documented to enhance the reaction rate. In the catalytic enantioselective opening of 3-phenyloxetane **2a** with TMSCl^{43–45} (Fig. 3a), previous investigations had shown that the application of equimolar loadings of achiral thiourea **1a** and chiral squaramide **1c** led to marked enhancement in rate relative to each catalyst alone, while not diminishing the enantioselectivity of the product obtained with only **1c** (refs. 46,47). In this validation study, we performed the reactions under more challenging and practical conditions: lower catalyst loading (1 mol%) and higher reaction temperature under ambient atmosphere (Fig. 3a). We chose a panel of $N = 11$ catalyst candidates **1a–1k**, drawing representatives from classes of known organocatalysts (Fig. 3a).

Pooling–deconvolution was initiated by measuring the yields of individual candidates at low conversion (3 h) (Fig. 3a). Simulations with $N = 11$ led to the selection of (k , t , r) = (4, 2, 1) as the optimal parameter set (see Supplementary Information section 3.5.2). With small N , simulations showed that redundancy ($r > 1$) was not required to achieve high sensitivity. This corresponds to a design of 11 pools of four catalysts each, which was then tested experimentally (Fig. 3b, step 1). At the pooling stage, pool 3 (candidates **1a**, **1c**, **1d** and **1g**) exhibited the highest cooperativity ($Q = 0.9$). On deconvolution, catalyst pairs (**1c** and **1d**) and (**1c** and **1g**) were found to have substantially lower D scores due to their presence in other low- Q pools (pools 7 and 5, respectively) (Fig. 3b, step 2). The four remaining pairs tied for the top score and were tested separately in the verification phase (Fig. 3b, step 3). The pair (**1a**, **1c**), previously known to be cooperative, emerged as the most cooperative ($Q = 2.4$), affording product **4** in 60% yield and 87% e.e., providing successful validation of the workflow. Under optimized conditions, this

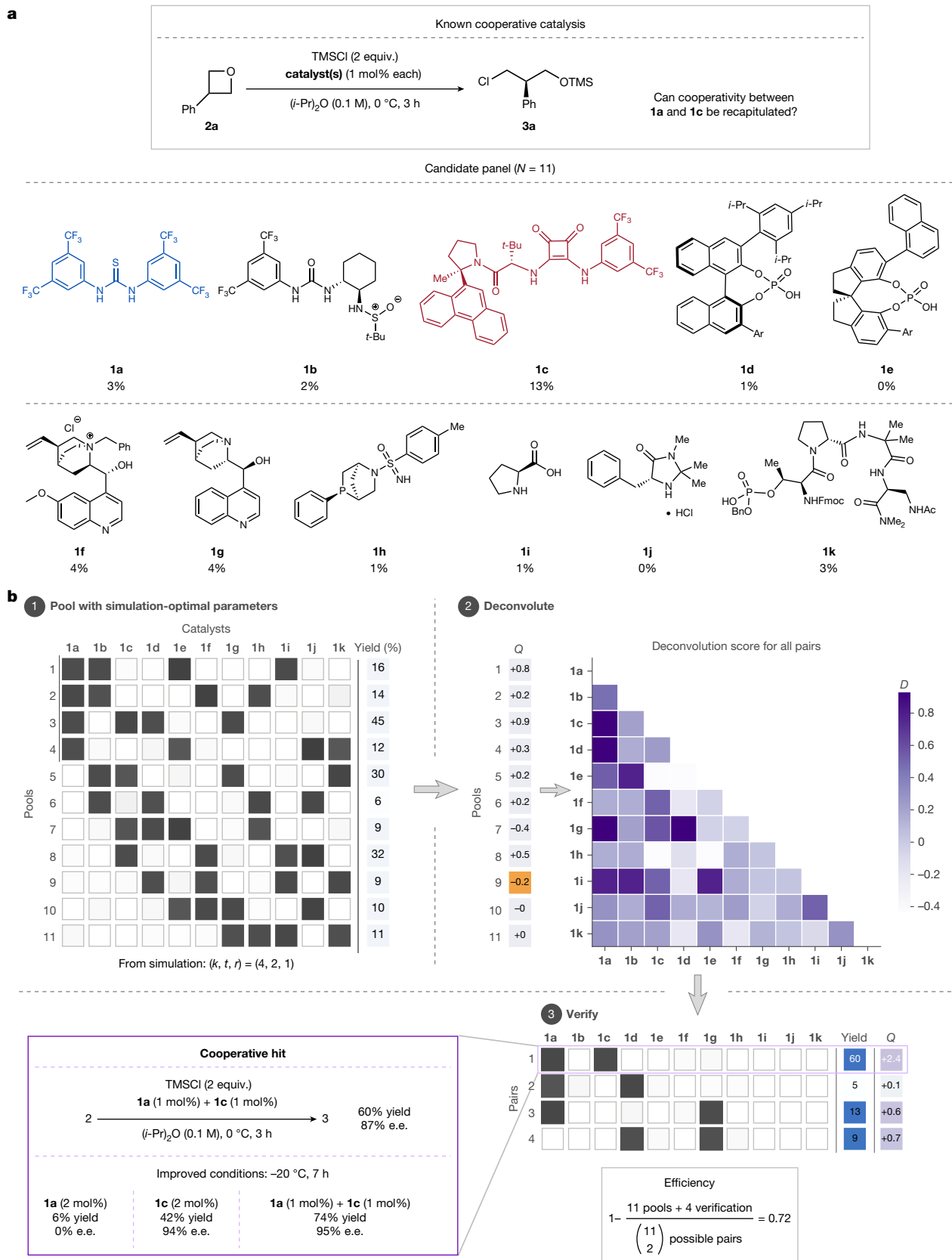


Fig. 3 | Validation of pooling–deconvolution workflow on a catalytic enantioselective oxetane opening. **a**, A chiral dual-hydrogen-bond-donor-catalysed opening of 3-phenyloxetane **2a** with TMSCl as the validation reaction and a panel of 11 candidates comprising representatives from privileged organocatalyst classes. Percentages shown are yields of **3a** determined by NMR analysis. **b**, Implementation of pooling–deconvolution for the oxetane

opening reaction. (1) Pool design with simulation-identified parameters (*k*, *t*, *r*) = (4, 2, 1). In each pool (row), filled squares represent the inclusion of a candidate. (2) Deconvolution scores for every candidate pair based on the performance of pools and individuals. (3) Individual verification of four pairs exhibiting the highest deconvolution scores, recapitulating the expected cooperative pair of **1a** and **1c**.

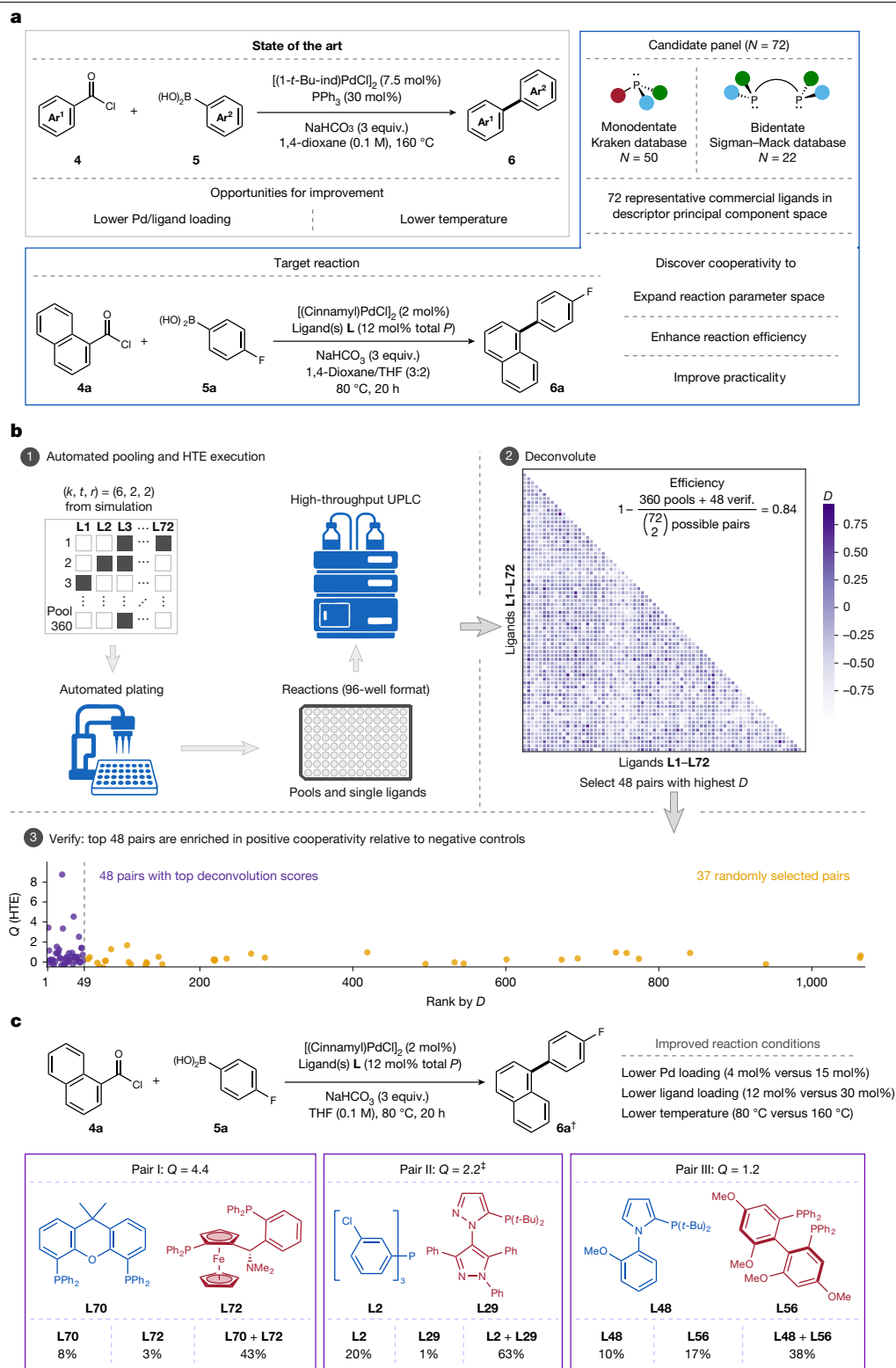


Fig. 4 | Deploying the pooling–deconvolution workflow to discover ligand cooperativity in a Pd-catalysed decarbonylative Suzuki–Miyaura coupling.

a, State-of-the-art conditions for the target cross-coupling transformation and diversity-oriented selection strategy for phosphine ligand candidate panel. **b**, Implementation of pooling–deconvolution. (1) Automated pooling and high-throughput experimentation/analysis workflow based on simulation-selected parameters. (2) Deconvolution scores for every candidate pair based on the performance of pools and individuals. (3) Verification of catalyst pairs by discrete testing. The y-axis, $Q(\text{HTE})$, is the cooperativity of a pair tested discretely relative to its constituent ligands in a small-scale, high-throughput setting. The x-axis of the

plot is the rank of each pair by deconvolution score D . The purple points are the 48 points with the highest calculated D , whereas the yellow points are a negative control group selected randomly from the remaining pairs. **c**, Validation of top cooperative hits in batch mode, with the three highest-yielding cooperative pairs shown. Q is calculated relative to the larger yield of the two individual ligands. The dagger symbol indicates yields were determined by quantitative ^{19}F NMR integration. The double dagger symbol indicates reaction was performed with 8 mol% total phosphorus atom instead of 12 mol% total phosphorus atom.

catalyst pair was shown to promote the reaction in 74% yield and 95% e.e. at 1 mol% loading of each catalyst, a substantial improvement in rate over the performance of the chiral catalyst alone (42% yield and 94% e.e. at 2 mol% **1c**).

The pooling–deconvolution process required 15 experiments to identify a cooperative pair in a space of 55 possible pairs, corresponding to an efficiency of 72%. No further hits with high positive cooperativity ($Q > 1.1$) were found on enumeration of every pair post facto (see Supplementary Information section 4.4), indicating that the workflow was perfectly accurate in this instance. Furthermore, these experiments showed that 31 of the 55 pairs exhibited weak to moderate negative cooperativity, reflecting the robustness of the pooling–deconvolution process to interference when negative cooperativity is common.

Pooling–deconvolution leads to discovery of new catalytic cooperativity

Next, we sought to apply this workflow in a discovery context, in which cooperativity was previously unknown. We selected the Pd-catalysed decarbonylative Suzuki–Miyaura coupling between aroyl chlorides **4** and arylboronic acids **5** towards biaryls **6** (Fig. 4a, top left), a transformation for which a practical solution remains highly desirable due to the abundance of available carboxylic acid derivatives^{48,49}. Existing catalytic systems require stepwise separation of the decarbonylation and biaryl coupling steps, high temperatures (130–160 °C), as well as high Pd loadings of 10–15 mol% (refs. 48,49). The harsh reaction conditions reflect the high kinetic barrier to decarbonylation as well as the inherent difficulty of integrating a challenging decarbonylation step into a Suzuki–Miyaura mechanistic sequence^{50–52}.

We proposed that a multi-ligand approach might be viable in which each ligand ostensibly facilitates a subset of the elementary steps⁵³. Guided by this motivation but without any more specific mechanistic hypothesis, we constructed a panel of 72 commercially available monodentate and bidentate phosphorus ligands by selecting diverse candidates with principal component analysis using previously published molecular descriptors^{54,55} (Fig. 4a, top right). The relative number of monodentate (50) and bidentate (22) ligands that ultimately comprise the ligand panel mirrors the relative number of commercially available ligands in each reference database. The cross-coupling between 1-naphthoyl chloride **4a** and 4-fluorophenylboronic acid **5a** was selected as a model reaction (Fig. 4a, bottom). To present a more stringent challenge for the discovery workflow, the catalyst loading and temperature were reduced to 4 mol% and 80 °C, respectively, both substantially lower relative to established protocols (Fig. 4a, bottom).

To select the optimal hyperparameter set for pooling $N = 72$ candidates, we performed custom simulations with $N = 72$ in a challenging hypothetical cooperativity scenario, which led to the identification of $(k, t, r) = (6, 2, 2)$ (see Supplementary Information section 3.5.3). These parameters correspond to a pooling design that evaluates 2556 possible ligand pairs with 360 pools of six ligands each. Although this experimental space could be in principle investigated manually, we opted for an automated, data-rich experimentation approach to maximize time and material efficiency. We used a high-throughput workflow comprising robotic liquid handling for ligand pool construction, reaction execution in a 96-well format at 10- μ mol scale, followed by ultrahigh-pressure liquid chromatography for yield quantitation (Fig. 4b, step 1). Defining Q as the enhancement in yield relative to the maximum yield of the six ligands in each pool, the resulting pool yields were deconvoluted into a set of 2,556 D scores (Fig. 4b, step 2). The 48 top-scoring pairs, together with 37 control pairs selected randomly from the remaining 2,508 pairs, were then tested experimentally. Although the D score does not correlate strictly with the degree of cooperativity of the individual pairs, the top-48 group was found to be enriched in highly cooperative

pairs (odds ratio = 9.5) relative to the control group⁵⁶ (Fig. 4b, step 3; see also Supplementary Information section 5.4.3). With 408 experiments performed relative to 2,556 ligand pairs, the experimental savings corresponded to 84%.

The top six cooperative pairs were validated successfully in preparative-scale batch reactions. Three of these, when used as ancillary ligand pairs without further optimization, afforded practically useful yields (Fig. 4c) at substantially reduced catalyst loadings and temperature compared with the state-of-the-art conditions (see Supplementary Information section 5.4.5). These ligand combinations, which appear unexpected and unlikely to be selected solely based on expert intuition, now serve as starting points for ongoing methodology development and mechanistic investigations that will be reported separately. These results indicate that ligand cooperativity in Pd catalysis can be leveraged to expand the functional space of any given set of ligands^{57–66} and that the empirical exploration of combinatorial ligand space with pooling–deconvolution is a valuable alternative to rational design or to serendipity^{6,8}.

Outlook

This work demonstrates that combining combinatorial pooling designs with simple deconvolution steps provides a systematic and experimentally tractable approach to discover catalytic cooperativity in an empirical manner. This discovery strategy is built on the well-established mathematics of covering designs and group testing without requiring any chemical information about the reaction or about each catalyst candidate. We anticipate that this approach will enable the identification of cooperative interactions whose mechanistic underpinnings may be unknown but are probably rich and complex, laying the foundation for further reaction development and mechanistic investigations. Furthermore, we foresee expanding this workflow to identify ternary and higher-order cooperativity using diverse catalyst panels, including mixtures of biocatalysts, organic molecules and transition-metal complexes—in which cooperative mechanisms may differ substantially from the examples explored in this study. Overall, we believe that this approach will broadly expand opportunities to discover and leverage catalytic cooperativity, ultimately enabling more efficient chemical processes.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09813-2>.

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Data availability

All data are available in the main text, in the SI, and on Zenodo (<https://doi.org/10.5281/zenodo.17316238>). Illustrations of “automated plating” and “high-throughput UPLC” in Fig. 4b were created in Adobe Illustrator.

Code availability

All code is available on Zenodo (<https://doi.org/10.5281/zenodo.17316238>). In addition to a persistent version on Zenodo, the Python library developed for simulation and execution is maintained on GitHub under the GPL 3.0 license (<https://github.com/msh-yi/multicat-data>).

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Additional information

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